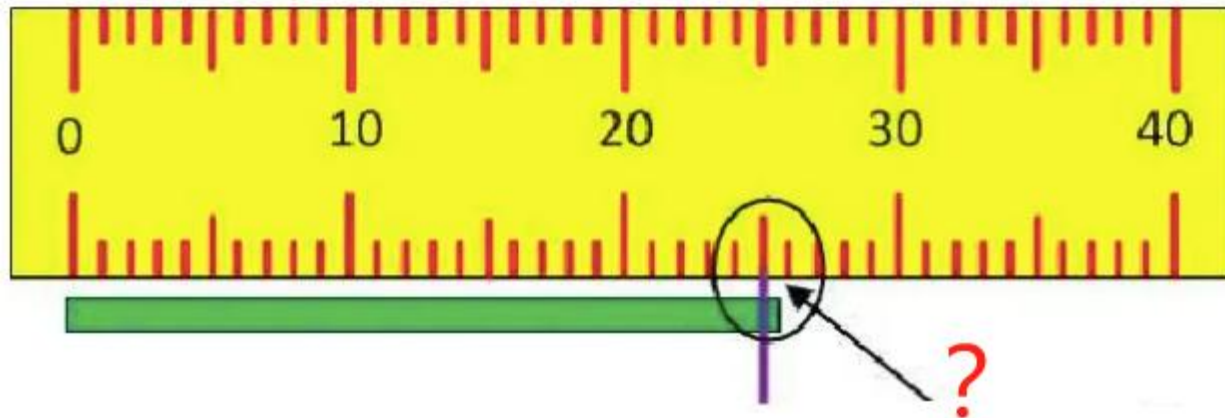
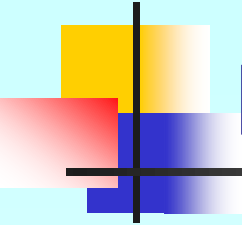


# Lecture 2. Measure and Spatial Calculation Based on GIS





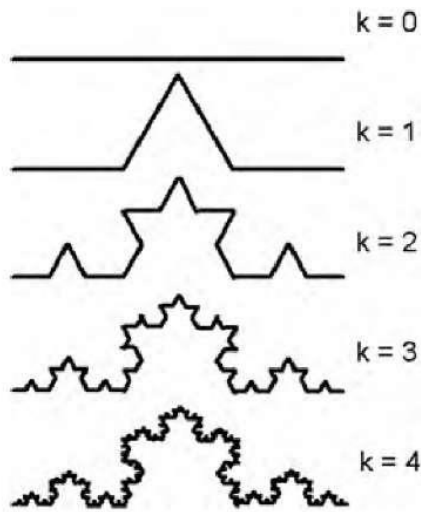
# Main Content of the Course

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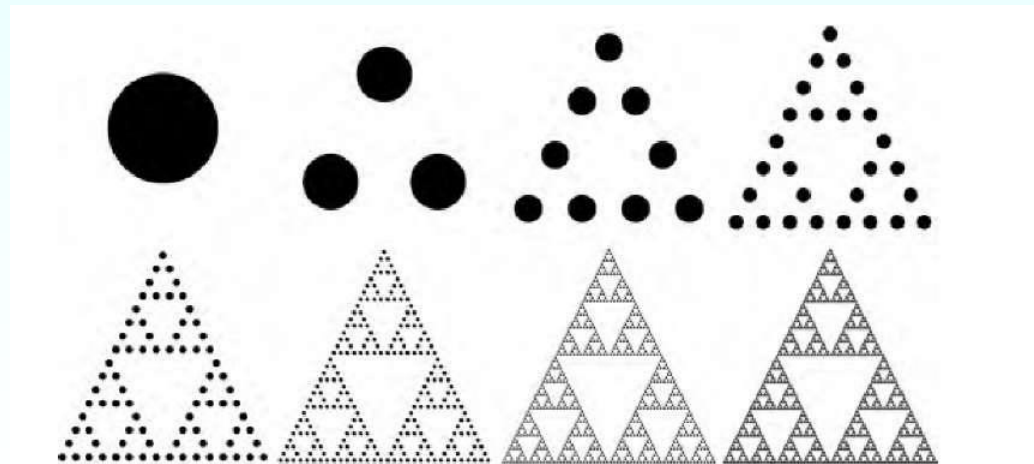
1. Dimension of the space
2. Measure: The Beginning of Spatial Analysis
3. Scale of Measure, Scale effect of spatial feature
4. Measuring the basic geometry parameters,  
Calculating the shape index, Calculating the  
spatial distribution
5. Error in Spatial Analysis

# 1. Dimension of the space

- Spatial Entity: 0, 1, 2, 3
- Spatial Phenomena: 2+T, 3+T
- Self-Similarity and The Fractal Dimension: **X.X**



Iterative steps to generate a  
Koch curve 科赫曲线

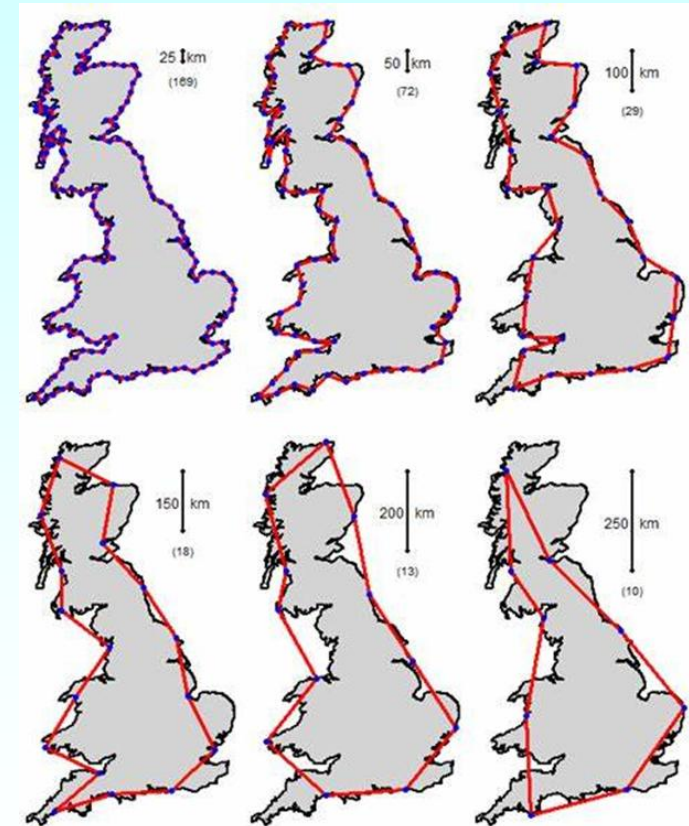


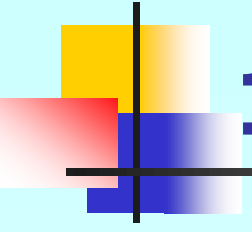
Iterative steps to construct the  
Sierpinsky gasket 谢尔平斯基镂垫

# 1. Dimension of the space



Coastline of Britain measurement  
with different length sticks





# 1. Dimension of the space

---

- 2 dimension

- Area, Perimeter, Center, Centroid

- 3 dimension

- Volume, Surface Area, Surface Perimeter

- 真三维GIS空间分析处理软件应该有什么功能?

## 2. Check and Measure: The Beginning of Spatial Analysis

### ■ Check Dataset

■ Georeference

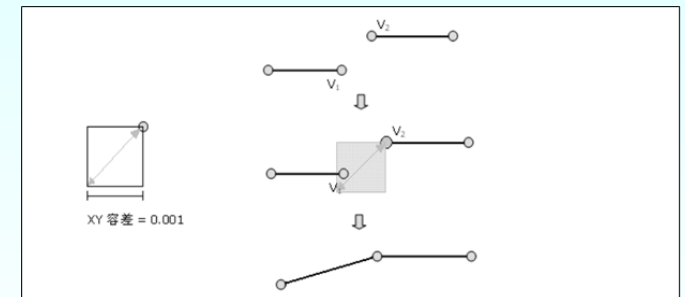
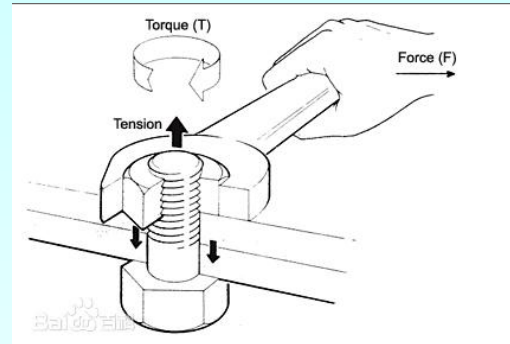
■ Unit

■ Scale

■ Tolerance (容差, 公差)

■ Domain

■ Object



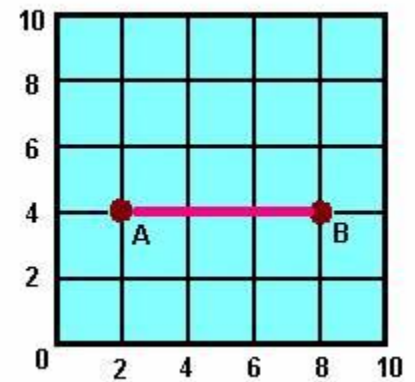
上: 3004443.38841  
下: 3001705.39017  
左: 13268516.8276  
右: 13271054.1786

?

上: 26.0593179086  
下: 26.0027240457  
左: 119.175895483  
右: 119.239226711

# 3. Scale of Measure

## 各种距离的定义



Distance in one dimensional space.

$$d(x, y) = \sqrt{(x_1 - y_1)^2} \quad (1) \quad (1)$$

Some frequently used distance functions.

Camberra :

$$d(x, y) = \sum_{i=1}^m \frac{|x_i - y_i|}{|x_i + y_i|} \quad (2)$$

Minkowsky :

$$d(x, y) = \left( \sum_{i=1}^m |x_i - y_i|^r \right)^{1/r} \quad (3)$$

Chebychev :

$$d(x, y) = \max_{i=1}^m |x_i - y_i| \quad (4)$$

Euclidean :

$$d(x, y) = \sqrt{\sum_{i=1}^m (x_i - y_i)^2} \quad (5) \quad (5)$$

Manhattan / city - block :

$$d(x, y) = \sum_{i=1}^m |x_i - y_i| \quad (6) \quad (6)$$

# 3. Scale of Measure

|                                |                                                                                                                                                                                                                                     |
|--------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Euclidean                      | $d(x, y) = \sqrt{\sum (x_i - y_i)^2}$                                                                                                                                                                                               |
| Squared Euclidean              | $d(x, y) = \sum (x_i - y_i)^2$                                                                                                                                                                                                      |
| Manhattan                      | $d(x, y) = \sum  x_i - y_i $                                                                                                                                                                                                        |
| Canberra                       | $d(x, y) = \sum \frac{ x_i - y_i }{ x_i + y_i }$                                                                                                                                                                                    |
| Chebychev                      | $d(x, y) = \max( x_i - y_i )$                                                                                                                                                                                                       |
| Bray Curtis                    | $d(x, y) = \frac{\sum  x_i - y_i }{\sum x_i + y_i}$                                                                                                                                                                                 |
| Cosine Correlation             | $d(x, y) = \frac{\sum (x_i y_i)}{\sqrt{\sum (x_i)^2} \sqrt{\sum (y_i)^2}}$                                                                                                                                                          |
| Pearson Correlation            | $d(x, y) = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum (y_i - \bar{y})^2} \sqrt{\sum (y_i - \bar{y})^2}}$                                                                                                                 |
| Uncentered Pearson Correlation | $d(x, y) = \frac{\sum x_i y_i}{\sqrt{\sum (y_i - \bar{y})^2} \sqrt{\sum (y_i - \bar{y})^2}}$                                                                                                                                        |
| Euclidean Nullweighted         | Same as Euclidean, but only the indexes where both x and y have a value (not NULL) are used, and the result is weighted by the number of values calculated. Nulls must be replaced by the missing value calculator (in dataloader). |



### 3. Scale effect of spatial feature

- Representation of rivers and roads: lines or polygons?
- Representation of lakes: points or polygons?



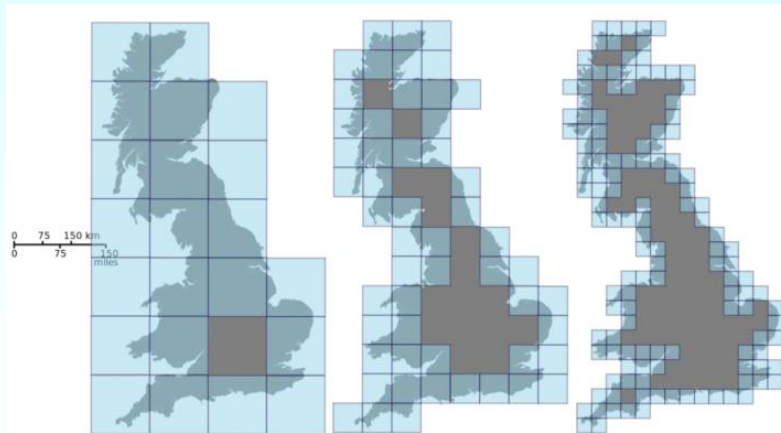
### 3. Scale of Measure

- Distance between: Points, Lines, polygons?



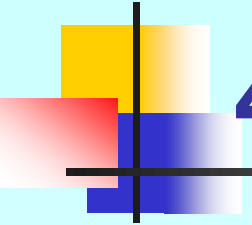
# 4. Measuring the basic geometry parameters

Length, Perimeter, Area, Center Coordinates  
Common Measures and Notation(de Smith, 2022)



| FID↵ | Name↵ | Area↵           |
|------|-------|-----------------|
| 0↵   | 黑龙江↵  | 415605.42680 ↵  |
| 1↵   | 新疆↵   | 1528345.87126 ↵ |
| 2↵   | 山西↵   | 148753.78240 ↵  |
| 3↵   | 宁夏↵   | 47811.35266 ↵   |
| 4↵   | 西藏↵   | 1189316.53864 ↵ |
| 5↵   | 山东↵   | 146827.73577 ↵  |
| 6↵   | 河南↵   | 159705.20943 ↵  |
| 7↵   | 江苏↵   | 97706.09394 ↵   |
| 8↵   | 安徽↵   | 138569.17819 ↵  |
| 9↵   | 湖北↵   | 184653.77875 ↵  |
| 10↵  | 浙江↵   | 101597.91439 ↵  |
| 11↵  | 江西↵   | 170568.53353 ↵  |
| 12↵  | 湖南↵   | 216628.17305 ↵  |
| 13↵  | 云南↵   | 404288.36043 ↵  |
| 14↵  | 贵州↵   | 181152.50134 ↵  |
| 15↵  | 福建↵   | 127067.63982 ↵  |
| 16↵  | 广西↵   | 250995.03787 ↵  |
| 17↵  | 广东↵   | 189046.50126 ↵  |

?



## 4. Calculating the shape index

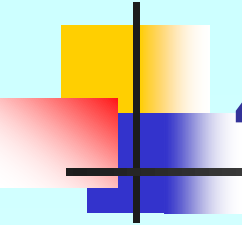
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### ■ Landscape index calculation

- 形状指数 Shape Index, 圆度指数 Circularity Index
- 几何中心 Geometric Center
- 质量重心 Center of Mass
- 外包络线 Outer Envelope Curve
- 外接圆 Circumcircle
- 外接矩形 Bounding Rectangle

### ■ Applications in Spatial query

**Measures of centrality** (de Smith, 2022)



## 4. Calculating the spatial distribution

---

### ■ Measures of distribution shape

- 景观破碎度 Landscape fragmentation degree
- 斑块特征 Mean patch area, number, density, shape index, .....
- 线密度（路网密度、河网密度） Line density (traffic density, river network density)
- 点密度 Point density

Measures of complexity and dimensionality (de Smith, 2022)

# 5. Error in Spatial Analysis

## ■ Error in Data and Measure

- SPOT-5: 2.5m?
- Systematic Errors, Random Errors, Gross Errors
- Nominal Error, Instrumental Errors, Environmental Errors, Observational Errors

## ■ The problem of error propagation in data processing/spatial analysis

- Area Calculation
- Buffer, Overlay
- Spatial Query with a buffer
- .....

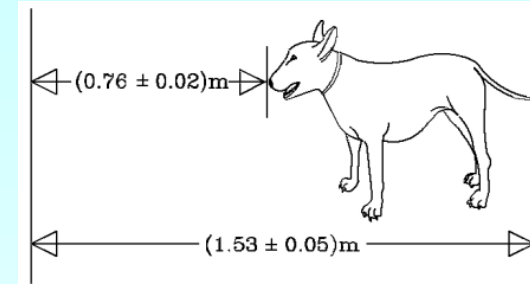
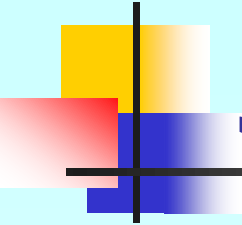


Figure 3: Measuring a Dog





## 5. Error in Spatial Analysis

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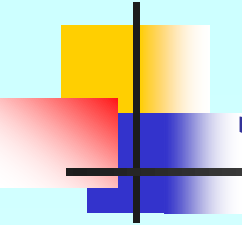
- 误差传播定律（Law of Error Propagation）本质上是基于多元函数的全微分和统计学中的方差性质推导出来的。
- 假设我们要计算一个物理量  $Z$ ，它是2个独立观测值  $x, y$  的函数：

$$Z = f(x, y)$$

- 在测量学中，我们假设观测误差  $\Delta x, \Delta y$  相比于测量值本身非常小。因此可以将函数  $f$  在测量值附近进行一阶泰勒展开（忽略高阶项）：

$$Z + \Delta Z \approx f(x, y) + \frac{\partial f}{\partial x} \Delta x + \frac{\partial f}{\partial y} \Delta y$$

$$\Delta Z \approx \frac{\partial f}{\partial x} \Delta x + \frac{\partial f}{\partial y} \Delta y$$



## 5. Error in Spatial Analysis

在实际应用中，单次测量的误差是随机的，我们更关心的是误差的统计特征——标准差 $\sigma$ 。

$$(\Delta Z)^2 = \left(\frac{\partial f}{\partial x} \Delta x\right)^2 + \left(\frac{\partial f}{\partial y} \Delta y\right)^2 + 2\left(\frac{\partial f}{\partial x}\right)\left(\frac{\partial f}{\partial y}\right)(\Delta x \Delta y)$$

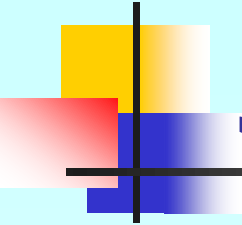
对等式两边取期望 (Expectation)。在统计学中：

- $E[(\Delta x)^2]$  就是变量  $x$  的方差  $\sigma_x^2$ 。
- 如果  $x$  和  $y$  是**相互独立**的观测值（例如你用 GPS 测长度和测宽度是两次独立操作），那么它们的协方差  $E[\Delta x \Delta y] = 0$ 。

$$\sigma_Z^2 = \left(\frac{\partial f}{\partial x}\right)^2 \sigma_x^2 + \left(\frac{\partial f}{\partial y}\right)^2 \sigma_y^2$$

$$\sigma_Z = \sqrt{\left(\frac{\partial f}{\partial x} \sigma_x\right)^2 + \left(\frac{\partial f}{\partial y} \sigma_y\right)^2}$$





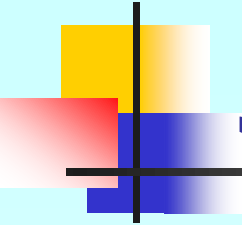
## 5. Error in Spatial Analysis

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在面积计算  $A = L \cdot W$  中:

1. 对  $L$  求偏导:  $\frac{\partial A}{\partial L} = W$
2. 对  $W$  求偏导:  $\frac{\partial A}{\partial W} = L$

$$\sigma_A = \sqrt{(W \cdot \sigma_L)^2 + (L \cdot \sigma_W)^2}$$



## 5. Error in Spatial Analysis

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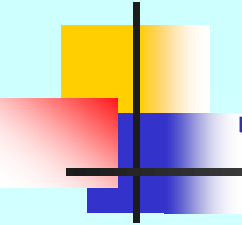
场景：计算一个矩形房屋地块的面积。

数据源：你使用皮尺测量地块的长和宽。仪器精度：皮尺的最小刻度是cm，测量误差是1.5cm。

测量值：长10.524 米，宽20.684米。

请计算面积，并给出含误差的表达。

- **测量仪器：**厘米级刻度皮尺（最小分度值 0.01 m）。
- **仪器中误差：**  $\sigma_L = \sigma_W = 0.015 \text{ m}$  (1.5 cm)。
- **长度观测值：**  $L = 10.524 \text{ m}$ （其中 0.004 为估读位，记录符合规范）。
- **宽度观测值：**  $W = 20.684 \text{ m}$ （其中 0.004 为估读位，记录符合规范）。



## 5. Error in Spatial Analysis

根据矩形面积公式  $A = L \times W$ :

$$A = 10.524 \times 20.684 = 217.678416 \text{ m}^2$$

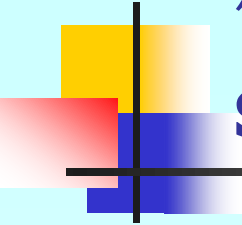
$$\sigma_A = \sqrt{\left(\frac{\partial A}{\partial L} \sigma_L\right)^2 + \left(\frac{\partial A}{\partial W} \sigma_W\right)^2} = \sqrt{(W \sigma_L)^2 + (L \sigma_W)^2}$$

代入数值:

$$\sigma_A = \sqrt{(20.684 \times 0.015)^2 + (10.524 \times 0.015)^2}$$

$$\sigma_A = \sqrt{0.09626 + 0.02492} \approx \sqrt{0.12118} \approx 0.35 \text{ m}^2$$

$$A = 217.7 \pm 0.4 \text{ m}^2$$



# 作业：Data preparation before spatial analysis

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## ■ 数据检查

- 投影与坐标系

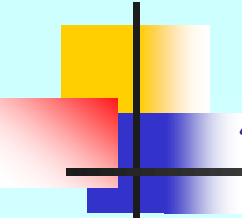
- 量测单位

## ■ 空间量算的误差分析

- 点的位置，线的长度

- 面的周长、面积、形状指数（选择一种）

- 以人口、**GDP**等专题数据为例制作专题图，注意制图规范



# 思考题

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1. 写出实验报告，对形状指数进行制图输出，分析其指示意义。制图的过程中注意专题图的规范。
2. 基于GIS软件进行空间量算的过程中，需要注意哪些技术细节？
3. 总结空间数据可能出现的所有空间参考问题。
4. 空间量测计算与数据分析过程中的误差分析